



มหาวิทยาลัยขอนแก่น

วิทยา จริยา ปัญญา

KHON KAEN UNIVERSITY



HTML5 (Part 1)

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Agenda

- What is HTML5?
- Why HTML5?
- HTML Basics
- Some HTML5 Elements and APIs



What is HTML5?

- HTML5 is a standard for structuring and presenting content on the World Wide Web
- It includes detailed processing models to encourage more interoperable implementations
- It extends, improves and rationalizes the markup available for documents, and introduces markup and APIs for complex web applications



Why HTML5?

- New elements such as `<video>`, `<audio>`, and `<canvas>` make it easy to include and handle multimedia and graphical content on the web
- New elements such as `<section>`, `<article>`, `<header>`, and `<nav>` are designed to enrich the semantic content of documents
- The APIs and DOM are no longer afterthoughts, but are fundamental parts of the HTML5 specification



New Features for a Modern Website

- New semantic elements
 - <header>, <footer>, and <section>
- Forms 2.0
 - New attributes have been introduced for <input>
- Persistent local storage
- WebSocket
 - A next-generation bidirectional communication technology for web applications



New Features for a Modern Website

- Canvas
 - Supports a two-dimensional drawing surface that you can program with JavaScript
- Audio & Video
 - Can embed audio or video on your web pages
- Geolocation
 - Now visitors can choose to share their physical location with your web application
- Drag and drop
 - Drag and drop the items from one location to another location on the same webpage



HTML5 DOCTYPE and character encoding

- The DOCTYPE declare for HTML5 is very simple

```
<!DOCTYPE html>
```

- The character encoding (charset) declaration is also very simple

```
<meta charset="UTF-8">
```



HTML Basic (1)

- All HTML documents must start with a document type declaration `<!DOCTYPE html>`
- The HTML document itself begins with `<html>` and ends with `</html>`
- The visible part of the HTML document is between `<body>` and `</body>`
- HTML headings are defined with `<h1>` to `<h6>` tags
- HTML paragraphs are defined with the `<p>` tag



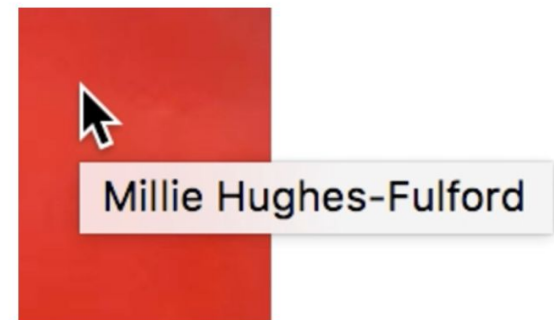
HTML Basic (2)

- HTML links are defined with the `<a>` tag (anchor)
 - Example: `<a href="http://www.kku.ac.th">KKU</a>`
- Links that open in new windows
 - Sometimes you want your links to open in new tabs or windows when users click them
 - To do so, add a second attribute to your `<a>` tag: the `target` attribute
 - By setting **target="_blank"**, you create a link that opens in a new window



HTML Basic (3)

- HTML images are defined with the `<img>` tag
 - The source file (src), alternative text (alt), and size (width and height) are provided as attributes
 - Example: ``
- The title of your image appears when a user hovers over the image itself



Hovering over an image will show contents of the title attribute



Stock images

Stock images are a safe choice when you're in a tight spot and need visuals to accompany your text!

Here are some solid options if you need stock photos:

- Adobe Stock (they often have free trials running): <https://stock.adobe.com>
- Shutterstock: <http://www.shutterstock.com/>
- iStock: <http://www.istockphoto.com/>
- Getty Images: <http://www.gettyimages.com/>
- Fotolia: <https://fotolia.com/>
- Picography (free, more niche/hipster): <https://picography.co>



Image formats

- **JPEG (Joint Photographic Experts Group)**
 - JPEG is a good format to use when adding a photo with many colors or with photography in general
- **GIF (Graphics Interchange Format)**
 - GIF is perfect for animated images. It's less suited to photography though because it can't handle as many colors
- **PNG (Portable Network Graphics)**
 - PNG allow transparency and do not result in quality loss when saved over and over again (unlike JPEGs)
- **SVG (Scalable Vector Graphics)**
 - SVG images are vector images. They can be scaled up or down and not lose any quality



Use Lowercase Tags

- HTML tags are not case sensitive: <P> means the same as <p>
- The HTML5 standard does not require lowercase tags
- But **W3C recommends lowercase in HTML, and demands lowercase for stricter document types like XHTML**



HTML Page Structure

```
<html>  
  
  <head>  
    <title>Page title</title>  
  </head>  
  
  <body>  
    <h1>This is a heading</h1>  
    <p>This is a paragraph.</p>  
    <p>This is another paragraph.</p>  
  </body>  
  
</html>
```



HTML lang Attribute

- Declaring a language is important for accessibility applications (screen readers) and search engine
- Example:
 - `<html lang="en-US">`
The first two letters specify the language (en). If there is a dialect, use two more letters (US).
 - `<html lang="th">`



HTML Style Attribute (1)

- Setting the style of an HTML element, can be done with style attribute
- `<tagname style="property:value;">`
- The property is a CSS property
- The value is a CSS value
- Example
- `<body style="background-color:powderblue;">`



HTML Style Attribute (2)

- `<h1 style="color:blue;">This is a heading</h1>`
- `<p style="font-family:courier;">This is a paragraph.</p>`
- `<h1 style="font-size:300%;">This is a heading</h1>`
- `<h1 style="text-align:center;">Centered Heading</h1>`



HTML Formatting

- `<b>` - Bold text
- `<strong>` - Important text
- `<i>` - Italic text
- `<em>` - Emphasized text
- `<mark>` - Marked text
- `<small>` - Small text
- `<del>` - Deleted text
- `<ins>` - Inserted text
- `<sub>` - Subscript text
- `<sup>` - Superscript text



Formatting Example

```
format.html x
html5 > format.html > html
1 <!DOCTYPE html>
2 <html>
3   <body>
4     <h2>HTML <mark>Formatting</mark></h2>
5     <p>My favorite color is <strong>blue</strong></p>
6     <p>My favorite fruit is <em>longans</em></p>
7   </body>
8 </html>
```

Go to Definition	F12
Go to References	Shift+F12
Peek	>
Find All References	Shift+Alt+F12
Rename Symbol	F2
Change All Occurrences	Ctrl+F2
Format Document	Shift+Alt+F
Cut	Ctrl+X
Copy	Ctrl+C
Paste	Ctrl+V
Open with Live Server	Alt+L Alt+O
Stop Live Server	Alt+L Alt+C
Command Palette...	Ctrl+Shift+P

← → ↻ ⓘ 127.0.0.1:5500/html5/format.html

HTML Formatting

My favorite color is **blue**

My favorite fruit is *longans*



HTML Colors

- In HTML, a color can be specified using a color name, an RGB value, or a HEX value



Background-color set by using red

Background-color set by using blue

```
<!DOCTYPE html><html><body>
```

```
<h2 style="background-color:red">Background-color set by using red</h2>
```

```
<h2 style="background-color:blue;color:white">Background-color set by  
using blue</h2></body></html>
```



HTML Colors Using RGB



Background-color set by using rgb(255,0,0)

Background-color set by using rgb(255,255,255)

```
<h2 style="background-color:rgb(255,0,0)">  
Background-color set by using rgb(255,0,0)  
</h2>
```

```
<h2  
style="background-color:rgb(0,0,255);color:rgb(255,255  
,255)">  
Background-color set by using rgb(255,255,255)  
</h2>
```



HTML Colors Using Hex

- In HTML, a color can also be specified using a hexadecimal value in the form: #RRGGBB, where RR (red), GG (green) and BB (blue) are hexadecimal values between 00 and FF (same as decimal 0-255).
- For example, #FF0000 is displayed as red, because red is set to its highest value (FF) and the others are set to the lowest value (00).



HTML Colors Using Hex



Background-color set by using #FF0000

Background-color set by using #0000FF



HTML Links – Create a Bookmark

- HTML bookmarks are used to allow readers to jump to specific parts of a Web page.
- Sample HTML Code
 - Create a bookmark with the id attribute:
`<h2 id="tips">Useful Tips Section</h2>`
 - Add a link to the bookmark from within the same page
`<a href="#tips">Visit the Useful Tips Section</a>`
 - Add a link to the bookmark from another page
`<a href="html_tips.html#tips">Visit the Useful Tips Section</a>`



HTML Table Tags

- Use the HTML **<table>** element to define a table
- Use the HTML **<tr>** element to define a table row
- Use the HTML **<td>** element to define a table data
- Use the HTML **<th>** element to define a table heading
- Use the HTML **<caption>** element to define a table caption



HTML Tables Tags Example

<> table.html U X

html5 > <> table.html > table > tr > td

```
1 <table>
2   <tr>
3     <th>Month</th>
4     <th>Savings</th>
5   </tr>
6   <tr>
7     <td>January</td>
8     <td>100</td>
9   </tr>
10  <tr>
11    <td>February</td>
12    <td>200</td>
13  </tr>
14  <tr>
15    <td>March</td>
16    <td>300</td>
17  </tr>
18  <caption>Monthly Savings in 2021</caption>
19 </table>
20
```

← → ↻ ⓘ 127.0.0.1:5500/html5/table.html

Monthly Savings
in 2021

Month	Savings
January	100
February	200
March	300



HTML Table Tags Properties

- Use the CSS **border** property to define a border
- Use the CSS **border-collapse** property to collapse cell borders
- Use the CSS **padding** property to add padding to cells
- Use the CSS **text-align** property to align cell text
- Use the CSS **border-spacing** property to set the spacing between cells
- Use the **colspan** attribute to make a cell span many columns
- Use the **rowspan** attribute to make a cell span many rows
- Use the **id** attribute to uniquely define one table



Border property

```
3 <head>
4   <style>
5     h1 {
6       border: 5px solid red;
7     }
8
9     h2 {
10      border: 4px dotted blue;
11    }
12    div {
13      border: double;
14    }
15  </style>
16 </head>
17 <body>
18   <h1>A heading with a solid red border</h1>
19   <h2>A heading with a dotted blue border</h2>
20   <div>A div element with a double border.</div>
21 </body>
```



border-collapse and border-spacing

```
<style>
  table, td, th {
    border: 1px solid black;
    border-collapse: separate
  }
</style>
```

```
<head>
  <style>
    table, td, th {
      border: 1px solid black;
      border-collapse: collapse
    }
  </style>
</head>
```

← → ↻ ⓘ 127.0

Monthly Savings
in 2021

Month	Savings
January	100
February	200
March	300

← → ↻ ⓘ 127.0

Monthly Savings
in 2021

Month	Savings
January	100
February	200
March	300



rowspan

```
<table>
  <tr>
    <th>Month</th>
    <th>Savings</th>
    <th>Savings for holiday!</th>
  </tr>
  <tr>
    <td>January</td>
    <td>$100</td>
    <td rowspan="2">$50</td>
  </tr>
  <tr>
    <td>February</td>
    <td>$80</td>
  </tr>
</table>
```

Month	Savings	Savings for holiday!
January	\$100	\$50
February	\$80	



HTML Lists

- Unordered HTML List
 - An unordered list starts with the `<ul>` tag
 - Each list item starts with the `<li>` tag
- Sample code

```
<h2>An unordered HTML list</h2>
```

```
<ul>
```

```
<li>Coffee</li>
```

```
<li>Tea</li>
```

```
<li>Milk</li>
```

```
</ul>
```



Unordered List with Disc Bullets

- Coffee
- Tea
- Milk



List Item Marker

- The CSS list-style-type property is used to define the style of the list item marker

Value	Description
disc	Sets the list item marker to a bullet (default)
circle	Sets the list item marker to a circle
square	Sets the list item marker to a square
none	The list items will not be marked



Sample Usage with List Item Marker

- Code

```
<ul style="list-style-type:circle">  
  <li>Coffee</li>  
  <li>Tea</li>  
  <li>Milk</li>  
</ul>
```

- HTML Page



Unordered List with Disc Bullets

- Coffee
- Tea
- Milk



Ordered HTML List

- An ordered list starts with the `<ol>` tag
- Each list item starts with the `<li>` tag
- The list items will be marked with numbers by default
- The Type attribute
- `type="1"`: numbers (default)
- `type="A"`: uppercase letters, `type="a"`,
- `type="I"`: uppercase roman numbers, `type="i"`



HTML Forms

- An HTML form contain form elements
- Form elements are different types of input elements, like text fields, checkboxes, radio buttons, submit buttons, and more
- The `<input>` Element (the most important)
 - `type="text"` Define a one-line text input field
 - `type="radio"` Define a radio button
 - `type="submit"` Define a submit button



HTML Input Element

- The <input> element is one of the most commonly used form controls
- The basic syntax is
`<input type="text" id="username" name="username" value="default value" required>`

•



Input Attribute type

- The **type** attribute specifies what kind of input control to display. Some common values include:
 - **text**: A single-line text input field
 - **password**: A single-line text field that masks the input
 - **radio**: A radio button
 - **checkbox**: A checkbox
 - **submit**: A submit button for the form
 - **email**: An input field for an email address
 - **number**: An input field for numeric input
 - **date**: An input field for a date



Input Attribute id

- The **id** attribute provides a unique identifier for the input element
- It's used to
 - Associate the input with a label (for accessibility)
 - Target the element with JavaScript or CSS
 - `<label for="user-name">Name:</label>`
 - `<input type="text" id="user-name" name="name">`



Input Attribute name

- The **name** attribute specifies the name of the input, which is used when the form data is submitted
- It's show the server identifies which piece of data is which
- Example

```
<input type="text" name="first_name">
```



Input Attribute value

- The value attribute specifies the initial value of the input field
- Its usage varies depending on the input type
- For text inputs, it sets the initial text in the field
- For radio buttons and checkboxes, it sets the value that is sent when the input is selected
- Examples:
 - `<input type="text" name="city" value="New York">`
 - `<input type="radio" name="gender" value="male"> Male`



The required keyword

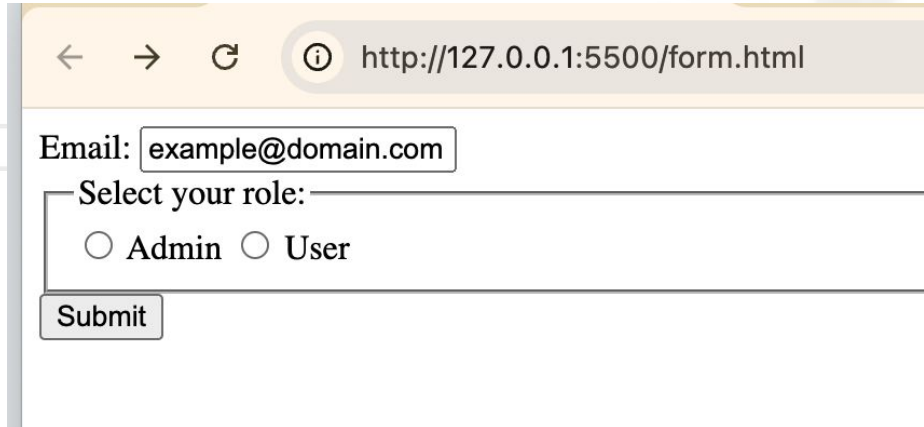
- The required keyword is a boolean attribute
- When present, it specifies the input field must be filled out before the form can be submitted
- Example:
 - `<input type="text" name="username" required>`



Example of Input Attributes

<> form.html >  form >  input#user-email

```
1  <form action="/submit" method="post">
2    <label for="user-email">Email:</label>
3    <input
4      type="email"
5      id="user-email"
6      name="email"
7      value="example@domain.com"
8      required
9    />
10
11  <fieldset>
12    <legend>Select your role:</legend>
13    <input type="radio" id="admin" name="role" value="admin" />
14    <label for="admin">Admin</label>
15    <input type="radio" id="user" name="role" value="user" />
16    <label for="user">User</label>
17  </fieldset>
18
19  <input type="submit" value="Submit" />
20 </form>
```



← → ↻ ⓘ http://127.0.0.1:5500/form.html

Email:

Select your role:

☐ Admin ☐ User



fieldset and legend

- Sample usage of fieldset and legend

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Personal Information

Name: Email:

← → ↻ ⓘ http://127.0.0.1:5500/form2.html ☆ 🧑 📊 🗑️ | 👤 Finish update ⋮

Contact Details

Name: Email:

Shipping Address

Street: City: ZIP Code:



fieldset

- The `<fieldset>` element is used to group related form controls together
- It's especially useful for longer forms where you want to organize inputs into logical sections.
- Key points about fieldset:
 1. It creates a box around the related elements.
 2. It helps improve form accessibility by grouping related form controls.
 3. It can be styled using CSS to visually separate different sections of a form.



legend

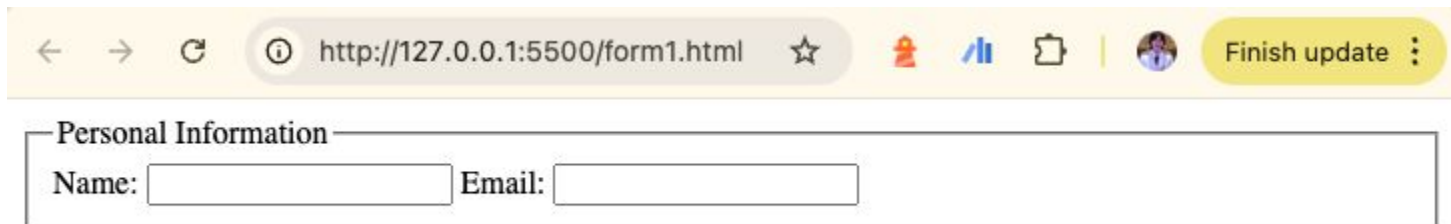
- The `<legend>` element is used in conjunction with `<fieldset>` to provide a caption or title for the grouped form controls.
- Key points about legend:
 1. It should be the first child of a fieldset.
 2. It provides a description for the group of form controls.
 3. Screen readers use the legend to describe the purpose of the fieldset to users.



Basic Usage of fieldset and legend

<> form1.html > ...

```
1  <form>
2    <fieldset>
3      <legend>Personal Information</legend>
4      <label for="name">Name:</label>
5      <input type="text" id="name" name="name" />
6      <label for="email">Email:</label>
7      <input type="email" id="email" name="email" />
8    </fieldset>
9  </form>
```



Personal Information

Name: Email:



Multiple Fieldsets in a Form

<> form2.html > form > fieldset > legend

```
1  <form>
2    <fieldset>
3      <legend>Contact Details</legend>
4      <label for="name">Name:</label>
5      <input type="text" id="name" name="name" />
6      <label for="email">Email:</label>
7      <input type="email" id="email" name="email" />
8    </fieldset>
9
10   <fieldset>
11     <legend>Shipping Address</legend>
12     <label for="street">Street:</label>
13     <input type="text" id="street" name="street" />
14     <label for="city">City:</label>
15     <input type="text" id="city" name="city" />
16     <label for="zip">ZIP Code:</label>
17     <input type="text" id="zip" name="zip" />
18   </fieldset>
19 </form>
```



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Contact Details

Name: Email:

Shipping Address

Street: City: ZIP Code:



Fieldset with Radio Buttons

<> form3.html >  form

```
1  <form>
2    <fieldset>
3      <legend>Select your preferred contact method:</legend>
4      <input type="radio" id="email" name="contact" value="email" />
5      <label for="email">Email</label><br />
6      <input type="radio" id="phone" name="contact" value="phone" />
7      <label for="phone">Phone</label><br />
8      <input type="radio" id="mail" name="contact" value="mail" />
9      <label for="mail">Physical Mail</label>
10   </fieldset>
11  </form>
```



← → ↻ ⓘ http://12... ☆   Finish update ⋮

Select your preferred contact method: _____

☐ Email

☐ Phone

☐ Physical Mail



Nested Fieldsets

<> form4.html > form > fieldset > fieldset > legend

```
1  <form>
2    <fieldset>
3      <legend>Your Feedback</legend>
4      <label for="name">Name:</label>
5      <input type="text" id="name" name="name" />
6
7    <fieldset>
8      <legend>Rate our service:</legend>
9      <input type="radio" id="excellent" name="rating"
10     value="excellent" />
11      <label for="excellent">Excellent</label><br />
12      <input type="radio" id="good" name="rating" value="good" />
13      <label for="good">Good</label><br />
14      <input type="radio" id="average" name="rating"
15     value="average" />
16      <label for="average">Average</label>
17    </fieldset>
18
19    <label for="comments">Additional Comments:</label>
20    <textarea id="comments" name="comments"></textarea>
21  </form>
```

← → ↻ ⓘ http://127.0.0.1:5500/form4.html

Your Feedback

Name:

Rate our service:

☐ Excellent

☐ Good

☐ Average

Additional Comments:



The Action Attribute

- The action attribute defines the action to be performed when the form is submitted
- Normally, the form data is sent to a web page on the server when the user clicks on the submit button

- Sample

```
<form action="action_page.php">
```

- If the action attribute is omitted, the action is set to the current page



The Method Attribute

- The method attribute specifies the HTTP method (GET or POST) to be used when submitting the form data
- Sample

```
<form action="action_page.php"  
method="get">
```

```
<form action="action_page.php"  
method="post">
```



When to Use GET?

- The default method is when submitting form data is GET
- However, when GET is used, the submitted form data will be visible in the page address field

```
action_page.php?firstname=Mickey&lastname=Mouse
```

- GET must NOT be used when sending sensitive information! GET is best suited for short, non-sensitive, amounts of data, because it has size limitations too



When to Use POST?

- Always use POST if the form data contains sensitive or personal information
- The POST method does not display the submitted form data in the page address field
- POST has no size limitations, and can be used to send large amounts of data



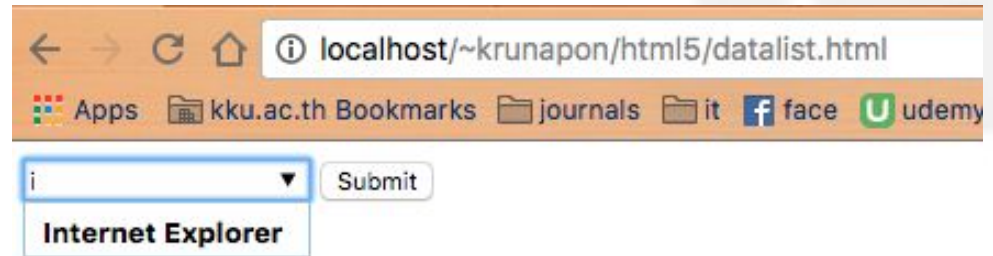
The Name Attribute

- Each input field must have a name attribute to be submitted
- If the name attribute is omitted, the data of that input field will not be sent at all



HTML5 <datalist> Element

- The **list** attribute of the <input> element, must refer to the **id** attribute of the <datalist> element



- Sample code

```
<form action="action_page.php">
  <input list="browsers" name="browser">
  <datalist id="browsers">
    <option value="Internet Explorer">
    <option value="Firefox">
  </datalist>
  <input type="submit">
</form>
```



<select> vs <datalist>

- **<select>** and **<datalist>** both let users pick from a list of options, but they behave differently:

```
<select name="browser">  
  <option  
value="chrome">Chrome</option>  
  <option  
value="firefox">Firefox</option>  
  <option  
value="safari">Safari</option>  
</select>
```

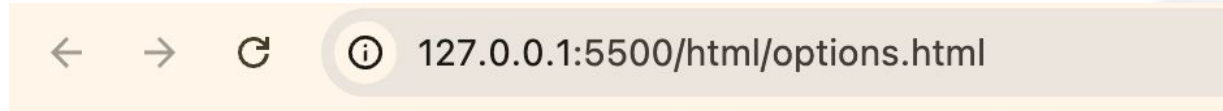
→ User must choose one of the three. No free typing.

```
<input list="browsers"  
name="browser">  
<datalist id="browsers"> <option  
value="Chrome"> <option  
value="Firefox"> <option  
value="Safari">  
</datalist>
```

→ User sees suggestions when typing, but can also type anything else (e.g. "Opera").



The Output of Using `<select>`



Options Using select

User must choose one of the choices. No free typing.



Using datalist

User can type or choose from the list.



The Output of Using <datalist>

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Options Using select

User must choose one of the choices. No free typing.

Chrome ▾

Options Using datalist

User can type or choose from the list.

Chrome

Firefox

Safari

← → ↻ ⓘ 127.0.0.1:5500/html/options.html

Options Using select

User must choose one of the choices. No free typing.

Chrome ▾

Options Using datalist

User can type or choose from the list.

Edge ▾



HTML5 <output> Element

- The <output> element represents the result of a calculation (like one performed by a script)

<form

oninput="x.value=parseInt(a.value)+parseInt(b.value)">

<input type="range" min="0" max="100" step="10" id="a" name="a" value="50">

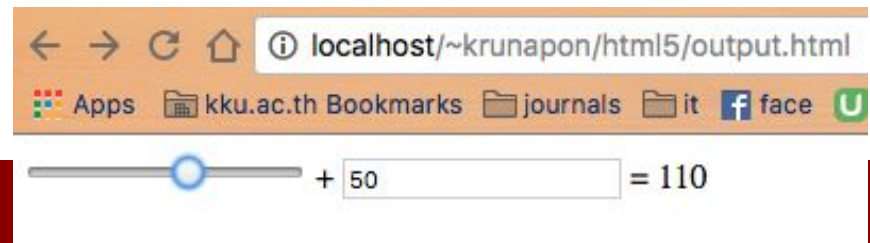
+

<input type="number" id="b" name="b" value="50">

=

<output name="x" for="a b"></output>

</form>



localhost/~krunapon/html5/output.html

Apps kku.ac.th Bookmarks journals it face U

+ 50 = 110



New HTML5 Elements

- The most interesting new HTML5 elements
 - New semantic elements like <header>, <footer>, <article>, and <section>
 - New attributes of form elements like number, date, time, calendar, and range
 - New graphic elements: <svg> and <canvas>
 - New multimedia elements: <audio> and <video>



Playing Videos in HTML

- The HTML5 <video> element specifies a standard way to embed a video in a web page

- Sample code

```
<video width="320" height="240" controls>  
  <source src="movie.mp4" type="video/mp4">  
  <source src="movie.ogv" type="video/ogg">  
  Your browser does not support the video tag.  
</video>
```



Playing Audio in HTML

- Sample code

```
<!DOCTYPE html>
```

```
<html>
```

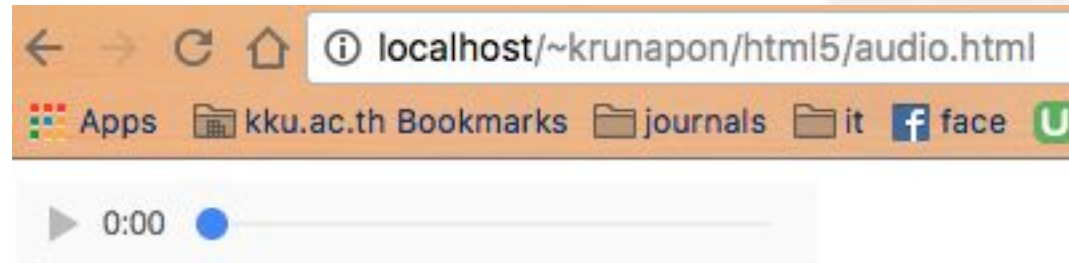
```
<body>
```

```
<audio controls>
```

```
<source src="horse.ogg" type="audio/ogg"> <source  
src="horse.mp3" type="audio/mpeg">Your browser does  
not support the audio element.</audio>
```

```
</body>
```

```
</html>
```



HTML YouTube Videos

- To play your video on a web page, do the following
 - Upload the video to YouTube
 - Take a note of the video id
 - Define an `<iframe>` element in your web page
 - Let the `src` attribute point to the video URL
 - Use the `width` and `height` attributes to specify the dimension of the player
 - Add any other parameters to the URL



Sample YouTube Code

- Sample code

```
<iframe width="420" height="345"  
src="https://www.youtube.com/embed/EerK  
1L0uXJ8"></iframe>
```



References

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- <http://www.tutorialspoint.com/html5/>
- <http://www.w3schools.com/html/default.asp>
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- <https://openclassrooms.com/en/courses/5265446-build-your-first-web-pages-with-html-and-css/5267576-add-images-to-your-web-page>
- <https://claude.ai>

